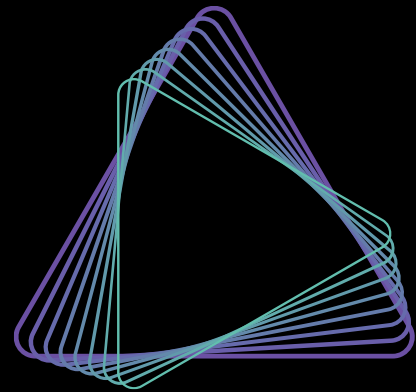




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LIFE SCIENCES

Seurat软件实际操作演练

转录组事业部

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数据准备

01

Seurat软件安装

Install Seurat v4

```
# Install the remotes package
if (!requireNamespace("remotes", quietly = TRUE)) {
  install.packages("remotes")
}
remotes::install_version("SeuratObject", "4.1.4", repos = c("https://satijalab.r-universe.dev",
getOption("repos")))
remotes::install_version("Seurat", "4.4.0", repos = c("https://satijalab.r-universe.dev",
getOption("repos")))
```


加载R包、读取数据及创建对象

#加载R包

```
library(Seurat)
```

```
library(dplyr)
```

```
library(SeuratData)
```

读取数据

```
data <- Read10X(data.dir = "path/to/your/data")
```

创建Seurat对象

```
object <- CreateSeuratObject(counts = data, project = "YourProjectName")
```

加载测试数据集

添加线粒体基因比例

```
library(SeuratData)
```

```
AvailableData()
```

```
object=UpdateSeuratObject(pbmc3k)
```

```
object
```

```
An object of class Seurat
13714 features across 2700 samples within 1 assay
Active assay: RNA (13714 features, 0 variable features)
2 layers present: counts, data
```

Seurat对象结构

一个典型的Seurat对象包含以下主要组件：

- **assays**：存储原始数据和各种处理后的数据。
- **meta.data**：存储与每个细胞相关的元数据，如细胞类型注释、QC指标等。
- **reductions**：存储降维结果，如PCA、t-SNE和UMAP。
- **graphs**：存储图结构数据，如邻接图，用于聚类分析。
- **neighbors**：存储最近邻计算结果。
- **commands**：记录Seurat对象处理过程中执行的命令。
- **tools**：存储各种分析工具生成的结果，如差异表达分析结果。

查看方式：object@

```
> object@
object@assays      object@active.ident  object@reductions  object@misc        object@tools
object@meta.data   object@graphs       object@images      object@version
object@active.assay object@neighbors    object@project.name object@commands
```


Assays介绍

assays是一个列表，包含一个或多个Assay对象。每个Assay对象代表一个数据集：

counts: 原始计数数据矩阵。

data: 标准化后的数据矩阵。

scale.data: 缩放后的数据矩阵。

var.features: 高变基因列表。

调用方式，以原始计数数据为例：

```
object@assays$RNA@counts
```


meta.data

- 细胞类型注释
- 质量控制指标（如nFeature_RNA、nCount_RNA、percent.mt等）
- 聚类结果

#查看

```
head(object@meta.data)
```

基本信息获取

`colnames(object)` #各个细胞的ID

`head(colnames(object))`

`Cells(object)` #各个细胞的ID, 同colnames

`head(Cells(object))`

`rownames(object)` # 基因名

`head(rownames(object))`

`ncol(object)` #列数，即细胞数

`nrow(object)` #行数，即基因数

`dim(object)` #查看对象的维度，即列数和行数

标准流程分析

02

细胞质量控制

添加线粒体基因比例

```
object[["percent.mt"]] <- PercentageFeatureSet(object, pattern = "^MT-")
```

基本统计信息的可视化

```
VlnPlot(object, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"), ncol = 3)
```

```
FeatureScatter(object, feature1 = "nCount_RNA", feature2 = "nFeature_RNA")
```

```
FeatureScatter(object, feature1 = "nCount_RNA", feature2 = "percent.mt")
```

筛选细胞

```
object <- subset(object, subset = nFeature_RNA > 200 & nFeature_RNA < 2500 & percent.mt < 5)
```


数据归一化、可变基因及数据标准化

```
object <- NormalizeData(object)
```

```
object <- FindVariableFeatures(object)
```

```
object <- ScaleData(object)
```

PCA降维

```
object <- RunPCA(object, features = VariableFeatures(object = object))
```

#PCA散点图

```
DimPlot(object, reduction = "pca")
```

可视化前两个主成分的基因负荷

```
VizDimLoadings(object, dims = 1:2, reduction = "pca")
```

绘制PCA热图，展示前20个主成分

```
DimHeatmap(object, dims = 1:10, cells = 100, balanced = TRUE)
```

绘制碎石图

```
ElbowPlot(object)
```

聚类分析

```
object <- FindNeighbors(object, dims = 1:15)
object <- FindClusters(object, resolution = 0.8)
#统计每个细胞亚群细胞数量
as.data.frame(table(Idents(object)))
```


UMAP/t-SNE可视化

UMAP:

```
object <- RunUMAP(object, dims = 1:10)  
DimPlot(object, reduction = "umap")
```

t-SNE:

```
object <- RunTSNE(object, dims = 1:10)  
DimPlot(object, reduction = "tsne")
```

#数据存储

```
object@reductions$
```

```
object@reductions$pca  object@reductions$umap  object@reductions$tsne
```


标志基因

识别某个亚群（如群体0）的标志基因

```
cluster0_markers <- FindMarkers(object, ident.1 = 0, min.pct = 0.25)
```

识别所有细胞亚群标志基因

```
all_markers <- FindAllMarkers(object, only.pos = TRUE, min.pct = 0.25,  
logfc.threshold = 0.25)
```

查看前几个标志基因

```
head(all_markers)
```


标志基因可视化

绘制标志基因在UMAP上的表达图

```
FeaturePlot(object, features = c("MS4A1", "CD79A"))
```

绘制标志基因的表达小提琴图

```
VlnPlot(object, features = c("MS4A1", "CD79A"))
```

绘制标志基因的热图

#选择top10基因

```
top10 <- all_markers %>% group_by(cluster) %>% top_n(n = 10, wt = avg_log2FC)
```

#绘图

```
DoHeatmap(object, features = top10$gene) + NoLegend()
```

SingleR自动注释细胞亚型

#加载R包

```
library(SingleR)
```

```
library(Seurat)
```

```
celltype <- "/home/rna015/Test/ref_Human_all.RData"
```

#加载事先下载参考数据集

```
ref.se <- get(load(celltype))
```

查看参考数据集

```
> ref.se
class: SummarizedExperiment
dim: 19363 713
metadata(0):
assays(1): logcounts
rownames(19363): A1BG A1BG-AS1 ... ZZEF1 ZZZ3
rowData names(0):
colnames(713): GSM112490 GSM112491 ... GSM92233 GSM92234
colData names(2): label.main label.fine
```

#细胞类型大类

ref.se\$label.main

#细胞类型细分

ref.se\$label.fine

SingleR注释

```
pred = SingleR(test = object@assays$RNA@data, ref = ref.se,  
               labels = ref.se$label.main,  
               clusters = object@active.ident,  
               assay.type.test = "logcounts", assay.type.ref = "logcounts")
```

pred

```
DataFrame with 9 rows and 4 columns  
      scores      labels delta.next pruned.labels  
      <matrix> <character> <numeric> <character>  
0 0.238018:0.583365:0.508006:... T_cells 0.0934376 T_cells  
1 0.241910:0.579114:0.500346:... T_cells 0.1051426 T_cells  
2 0.155920:0.503576:0.592474:... Monocyte 0.4801568 Monocyte  
3 0.210686:0.727590:0.538637:... B_cell 0.1072965 B_cell  
4 0.233986:0.568463:0.498143:... T_cells 0.2759478 T_cells  
5 0.172746:0.518636:0.570535:... Monocyte 0.4131160 Monocyte  
6 0.222100:0.530126:0.494991:... NK_cell 0.0676168 NK_cell  
7 0.181283:0.542873:0.549396:... Monocyte 0.2752186 Monocyte  
8 0.128344:0.276216:0.347441:... Platelets 0.2132783 Platelets
```


注释结果可视化

#提取注释到的细胞亚型

```
celltype = data.frame(ClusterID=rownames(pred),  
                      celltype=pred$labels, stringsAsFactors = F)
```

```
head(celltype)
```

#将注释结果添加进meta.data

```
object@meta.data$singleR=celltype[match(object@active.ident,celltype$ClusterID),'celltype']
```

#可视化

```
DimPlot(object, reduction = "umap", group.by = "singleR")
```

人工注释细胞亚型可视化

#查看当前的细胞聚类标签

`Idents(object)`

#将聚类标签映射到细胞类型，示例：

```
cluster_to_celltype <- c( "0" = "TypeA", "1" = "TypeB", "2" = "TypeC" )
```

#将SingleR注释到细胞类型进行人工映射

```
cluster_to_celltype <- c( "0" = "T_cells", "1" = "T_cells", "2" = "Monocyte ", "3" =  
"B_cells", "4" = "T_cells ", "5" = "T_cells ", "6" = "NK_cell ", "7" = "Monocyte ", "8" =  
"Platelets " )
```

#使用Renameldents重新标注细胞

```
object<- Renameldents(object, cluster_to_celltype)
```

```
Idents(object)
```

#可视化

```
DimPlot(object, reduction = "umap", label = TRUE) + NoLegend()
```


多样本整合

加载示例数据

```
merge=UpdateSeuratObject(ifnb)
```

```
ifnb.list <- SplitObject(merge, split.by = "stim")
```

```
ifnb.list
```

```
$CTRL
An object of class Seurat
14053 features across 6548 samples within 1 assay
Active assay: RNA (14053 features, 0 variable features)
 2 layers present: counts, data

$STIM
An object of class Seurat
14053 features across 7451 samples within 1 assay
Active assay: RNA (14053 features, 0 variable features)
 2 layers present: counts, data
```


#多样本整合分析时有两种处理方式：直接合并和批次效应校正后合并

#1. 直接合并

```
merged.ifnb<- merge(x = ifnb.list[[1]],y = ifnb.list[2])
```

#合并后进行后续标准分析

```
merged.ifnb <- NormalizeData(merged.ifnb)
```

```
merged.ifnb <- FindVariableFeatures(merged.ifnb, selection.method = "vst")
```

```
merged.ifnb <- ScaleData(merged.ifnb)
```

```
merged.ifnb <- RunPCA(merged.ifnb)
```

```
merged.ifnb <- FindNeighbors(merged.ifnb, dims = 1:30)
```

```
merged.ifnb <- FindClusters(merged.ifnb)
```

```
merged.ifnb <- RunUMAP(merged.ifnb, dims = 1:30)
```

#可视化

```
DimPlot(merged.ifnb, reduction = "umap", label = TRUE, repel = TRUE)
```

2. 批次效应矫正后合并

#对数据先进行标准化，并识别 variable feature

```
scelist <- lapply(X = ifnb.list , FUN = function(x) {  
  x <- NormalizeData(x)  
  x <- FindVariableFeatures(x, selection.method = "vst", nfeatures = 2000)  
})
```

#选择特征基因

```
features <- SelectIntegrationFeatures(object.list = scelist)
```

#查找锚点

```
immune.anchors <- FindIntegrationAnchors(object.list = scelist, anchor.features =  
features,reduction = "cca")
```

#数据整合

```
Inte<- IntegrateData(anchorset = immune.anchors)
```

```
DefaultAssay(Inte) <- "integrated"
```


#下游标准分析

#高变基因

```
Inte <- FindVariableFeatures(Inte,selection.method = "vst")
```

#数据标准化

```
Inte <- ScaleData(Inte, verbose = FALSE)
```

#降维及聚类

```
Inte <- RunPCA(Inte, features = VariableFeatures(object = Inte))
```

```
Inte <- FindNeighbors(Inte, reduction = "pca", dims = 1:15)
```

```
Inte <- FindClusters(Inte)
```

```
Inte <- RunTSNE(Inte, reduction = "pca", dims = 1:15)
```

```
Inte <- RunUMAP(Inte, reduction = "pca", dims = 1:15)
```


整合数据可视化

`ElbowPlot(Inte, ndims = 50) #PCA碎石图`

`DimPlot(Inte, reduction = "pca") #PCA图`

`DimPlot(Inte, reduction = "umap", group.by = "orig.ident") #按照样本标注细胞umap图`

`DimPlot(Inte, reduction = "umap", label = TRUE, repel = TRUE) #按照细胞分群标注细胞`

`DimPlot(Inte, reduction = "umap", split.by = "orig.ident") #按照样本拆分显示umap图`

标志基因

```
markers <- FindConservedMarkers(object, ident.1 = 0, grouping.var = "orig.ident",  
verbose = FALSE)  
  
head(markers)
```

保存及加载

#保存

```
save(object,file = "object.rData")
```

```
saveRDS(object,file = "object.rds")
```

#加载

```
load("object.rData")
```

```
readRDS("object.rds")
```


个性化分析

03

子集提取

#基于细胞类型/群体提取子集

```
B_cells <- subset(object, idents = "B_cells")
```

```
Cluster0_cells <- subset(object, idents = 0)
```

#基于基因表达水平提取子集

```
high_CD79_cells <- subset(object, subset = CD79A > 2)
```

#多条件组合提取子集

```
high_CD79_cells <- subset(object, subset = CD79A > 2, idents = " B_cells " )
```

细胞亚群再分群

#提取子集

```
B_cells <- subset(object, idents = "B cells")
```

#运行标准分析

```
B_cells <- FindVariableFeatures(B_cells)
```

```
B_cells <- ScaleData(B_cells)
```

```
B_cells <- RunPCA(B_cells)
```

```
B_cells <- FindNeighbors(B_cells, dims = 1:10)
```

```
B_cells <- FindClusters(B_cells, resolution = 0.5)
```

```
B_cells <- RunUMAP(B_cells, dims = 1:10)
```

#可视化

```
DimPlot(B_cells, reduction = "umap", label = TRUE, pt.size = 0.5) + NoLegend()
```

基因表达峰图

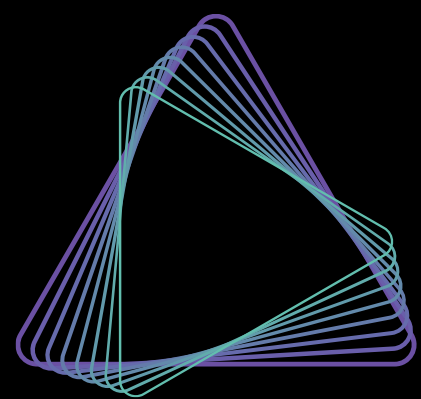
```
RidgePlot(object,features=c("CD79A","RPS3"),ncol=2)
```


基因表达气泡图

#选取基因

```
top2 <- all_markers %>% group_by(cluster) %>% top_n(n = 2, wt = avg_log2FC)  
DotPlot(object, features=top2$gene)+RotatedAxis()
```

Thank you!



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